

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

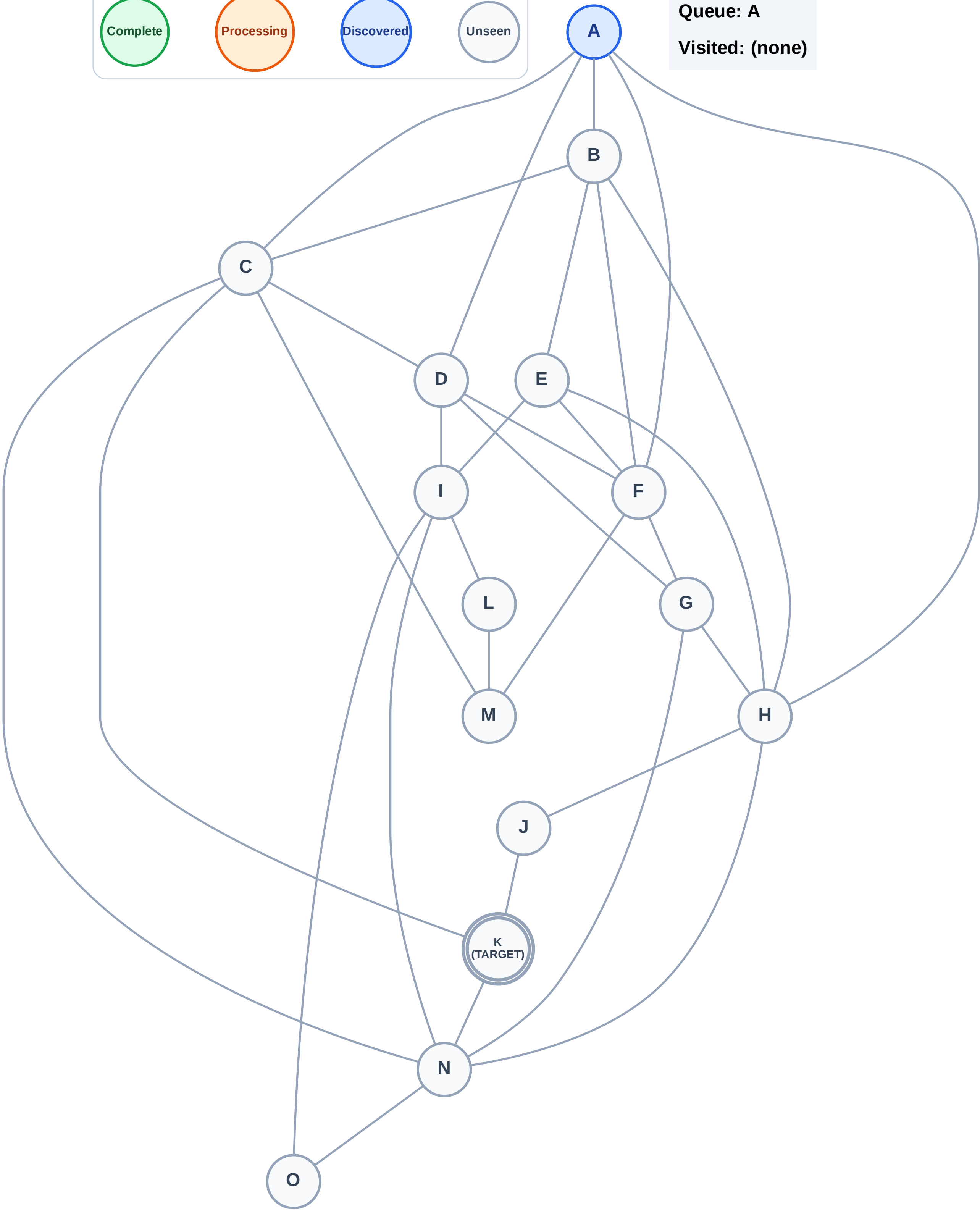
Processing

Discovered

Unseen

Queue: A

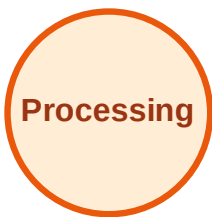
Visited: (none)



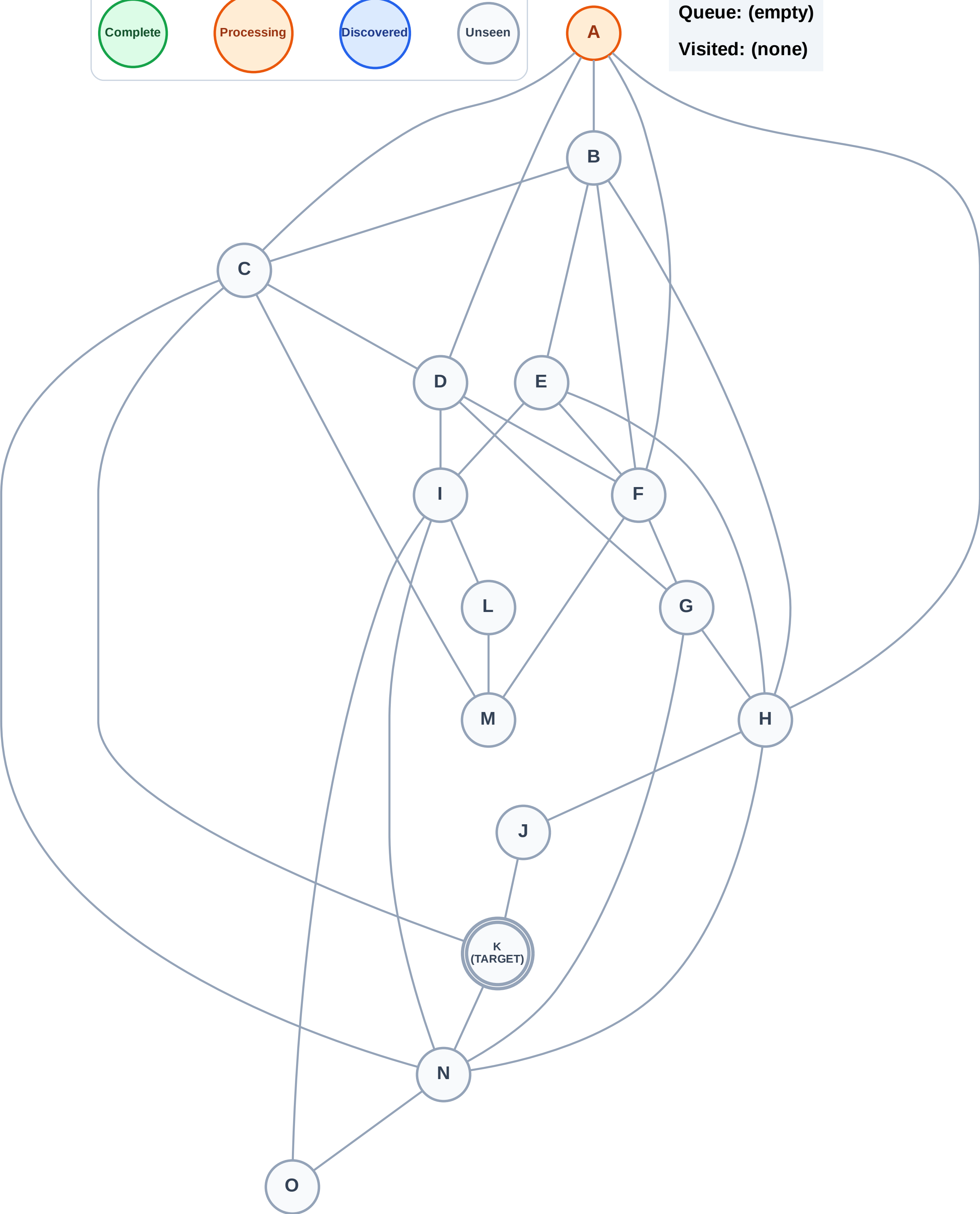
BFS Traversal

Step 2: Process node A

Legend



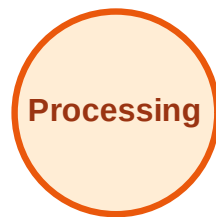
Queue: (empty)
Visited: (none)



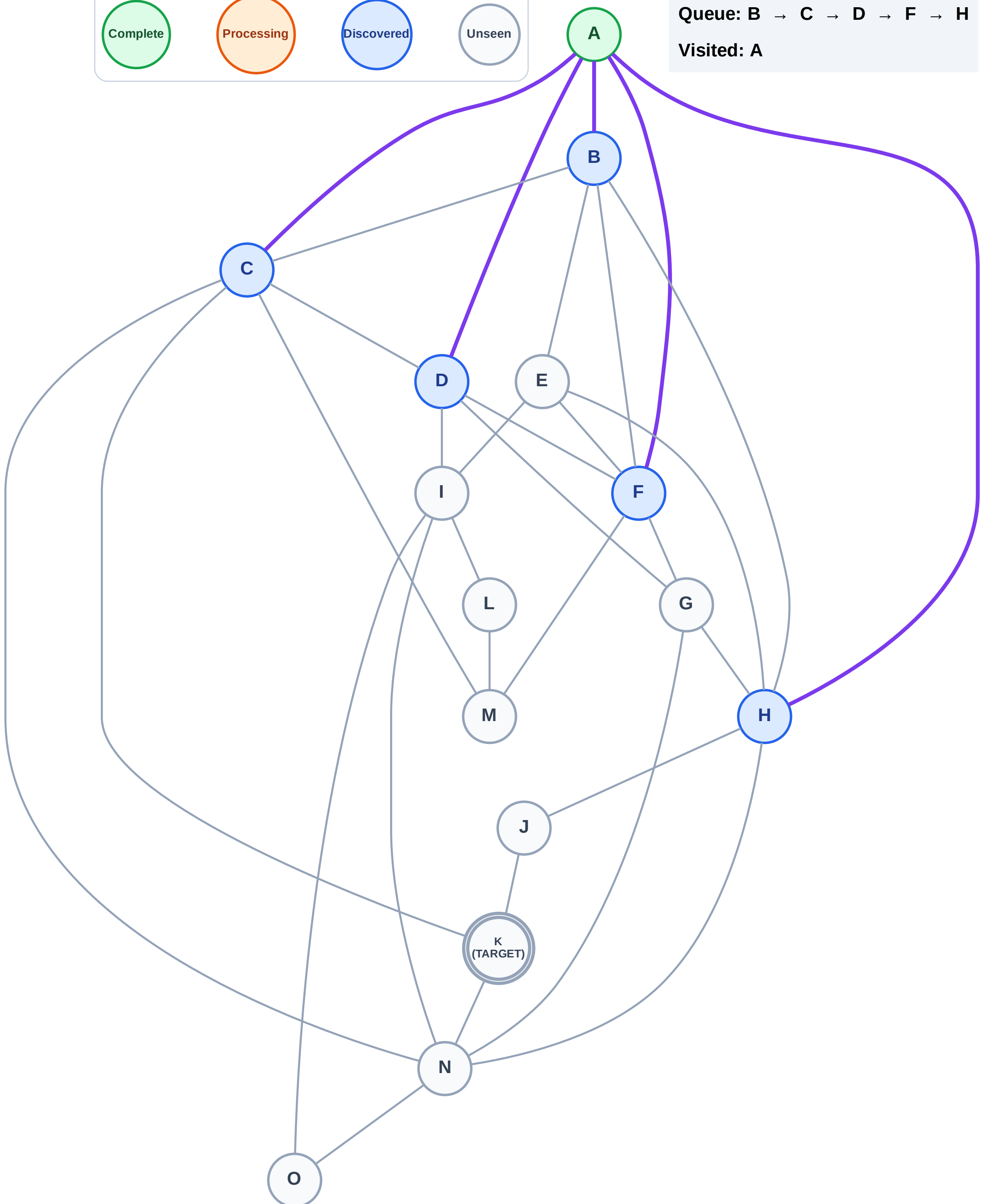
BFS Traversal

Step 3: Discover B, C, D, F, H from A

Legend



Queue: B → C → D → F → H
Visited: A



BFS Traversal

Step 4: Process node B

Legend

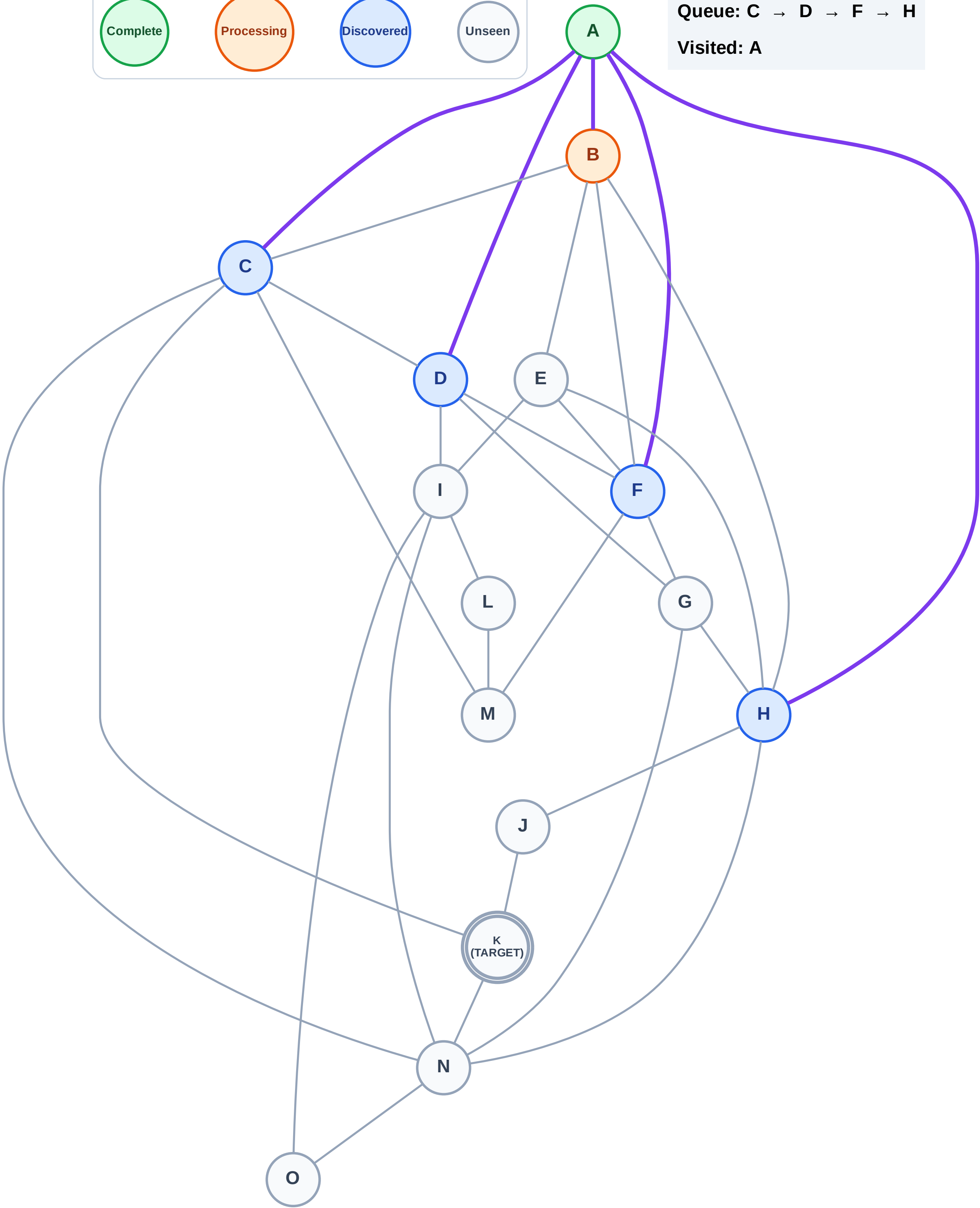
Complete

Processing

Discovered

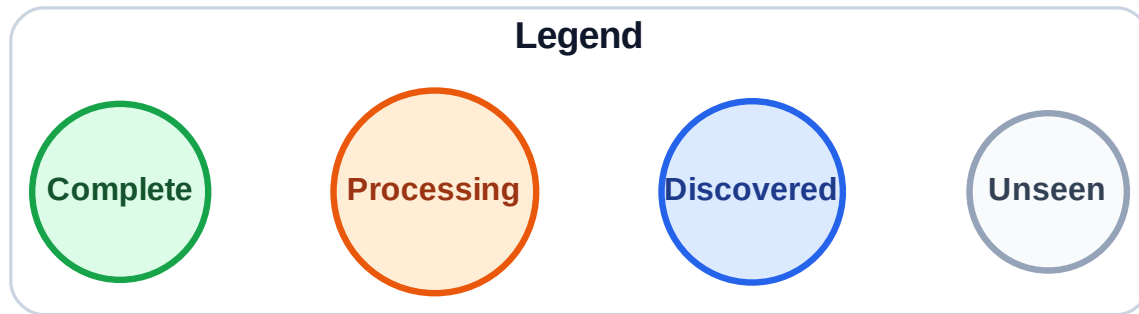
Unseen

Queue: C → D → F → H
Visited: A



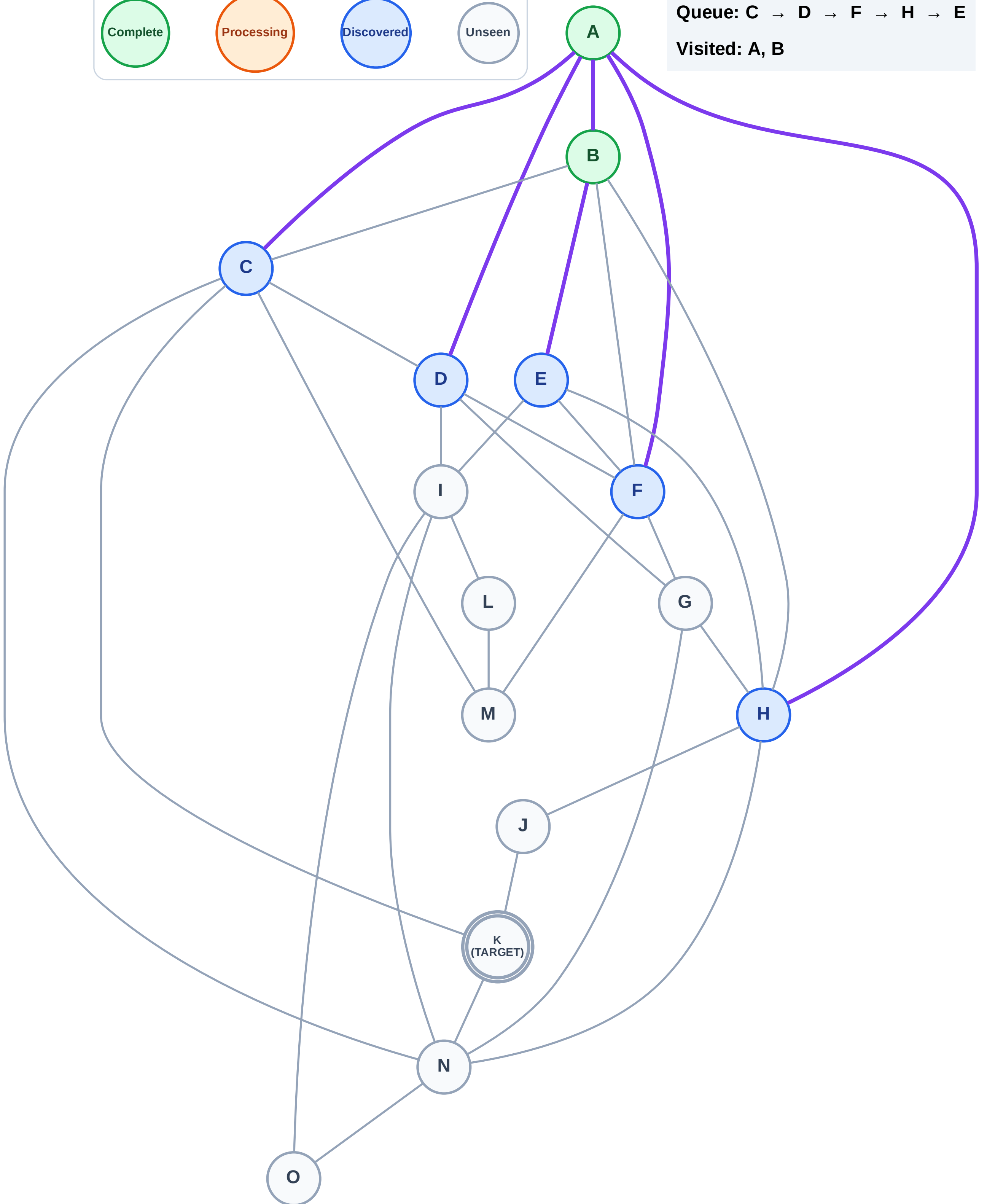
BFS Traversal

Step 5: Discover E from B



Queue: C → D → F → H → E

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

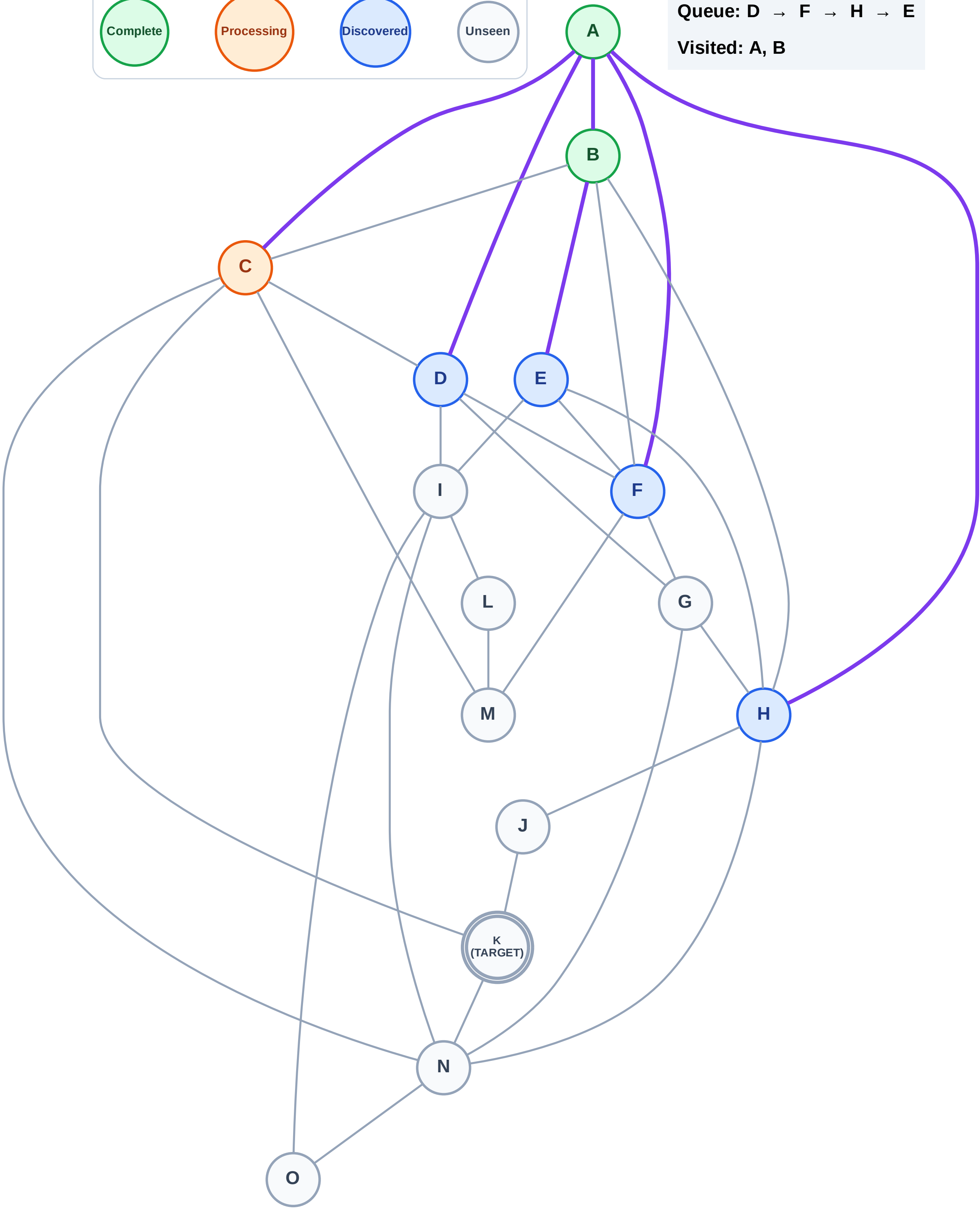
Processing

Discovered

Unseen

Queue: D → F → H → E

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend

Complete

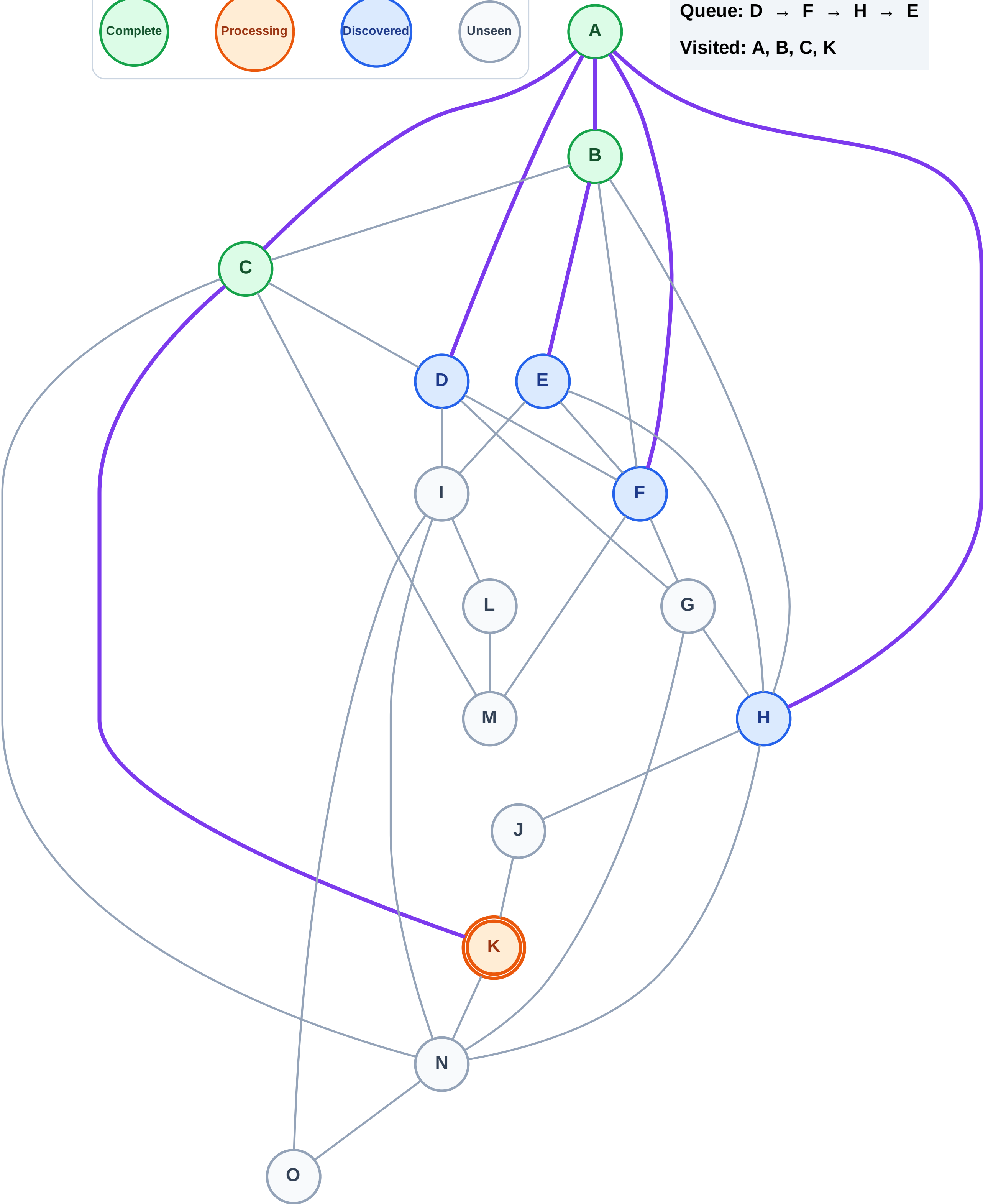
Processing

Discovered

Unseen

Queue: D → F → H → E

Visited: A, B, C, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

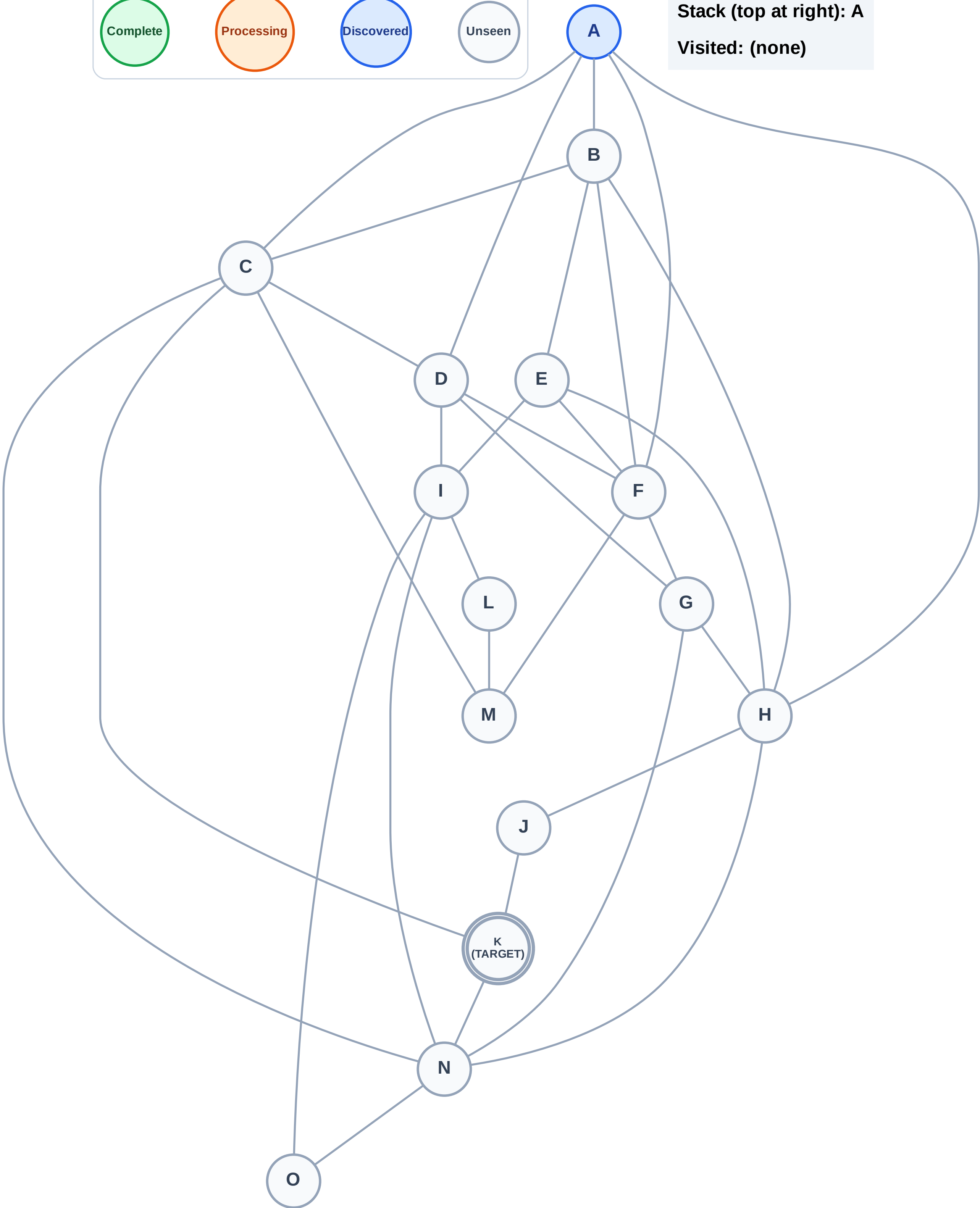
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

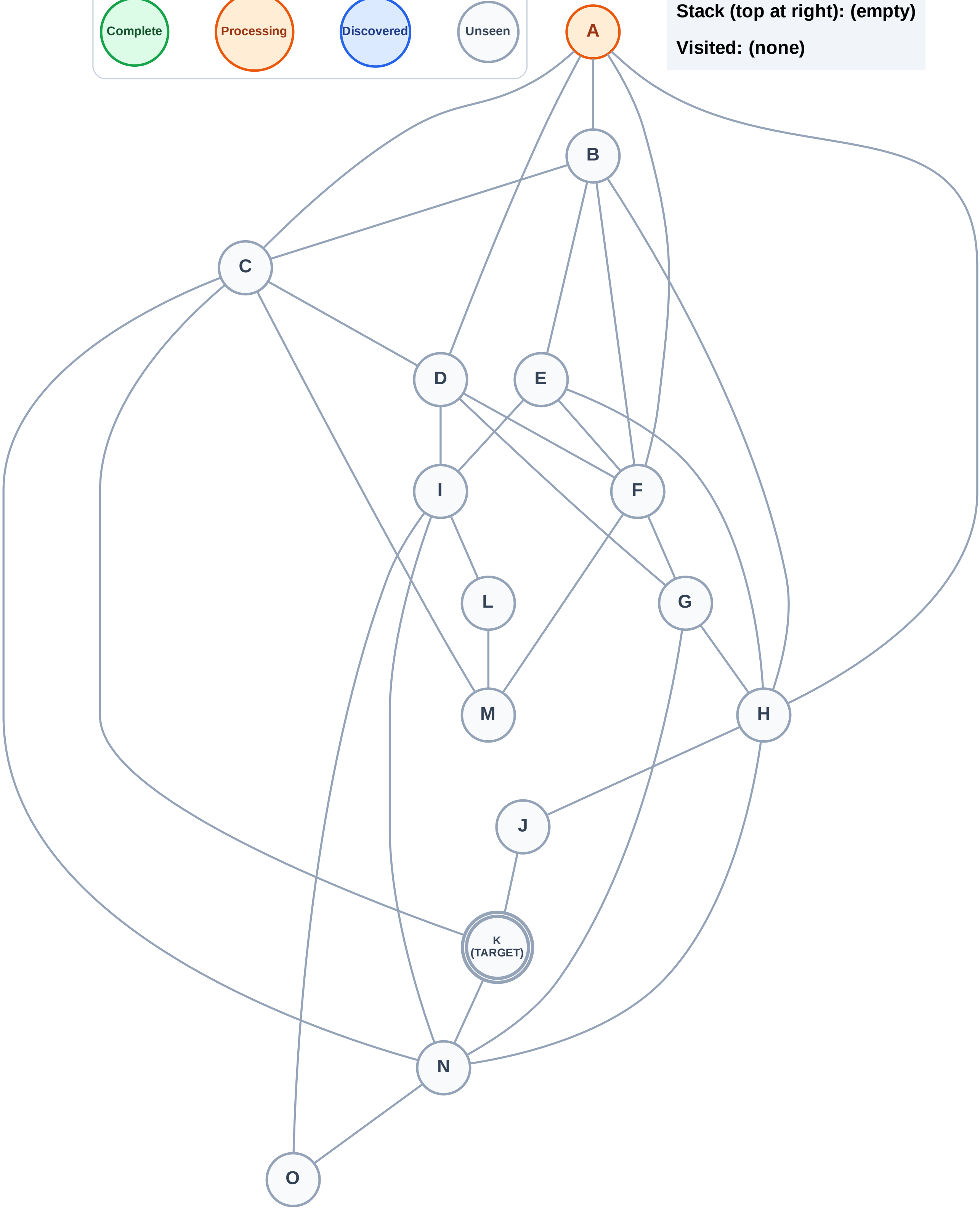
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, F, D, C, B from A

Legend

Complete

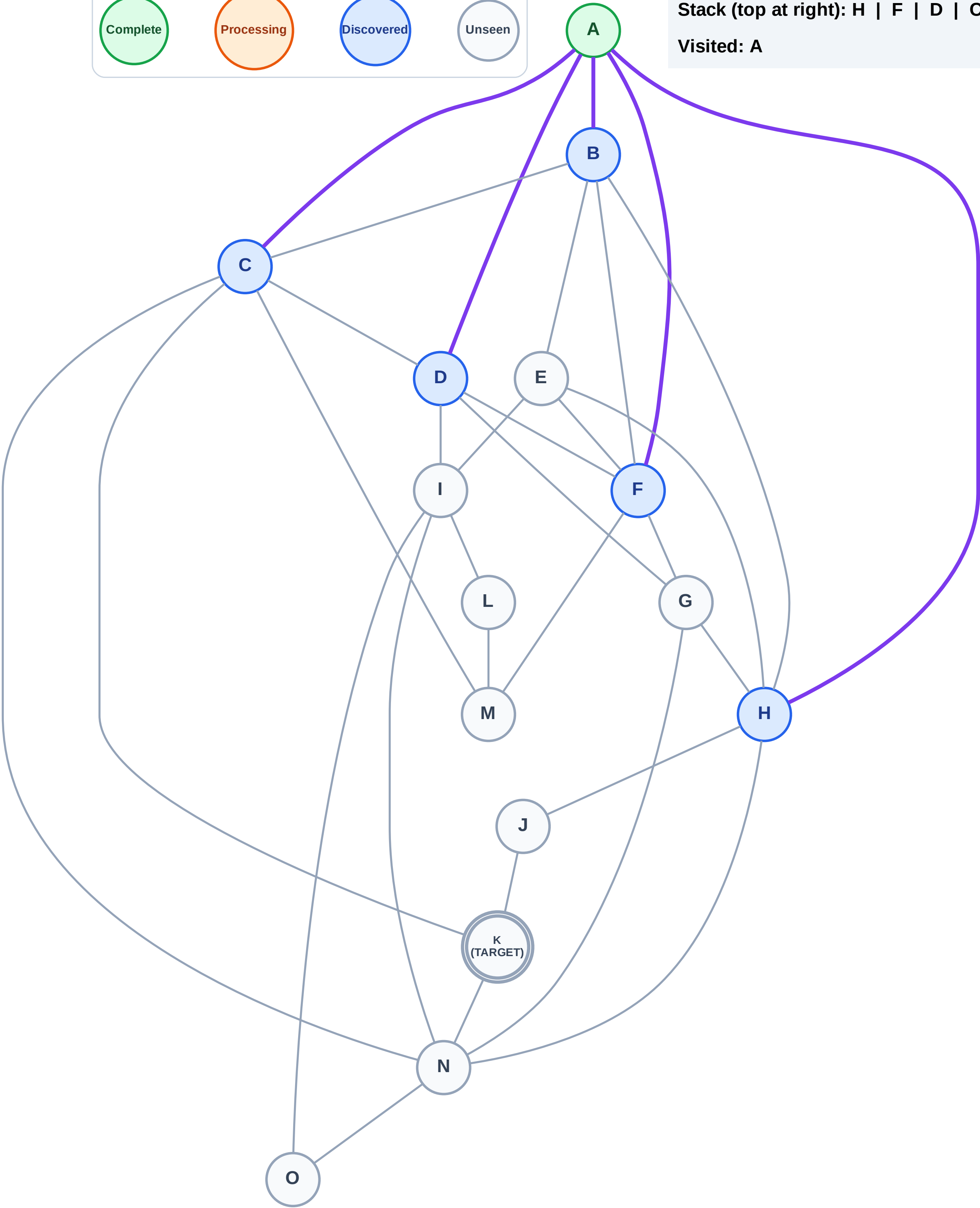
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

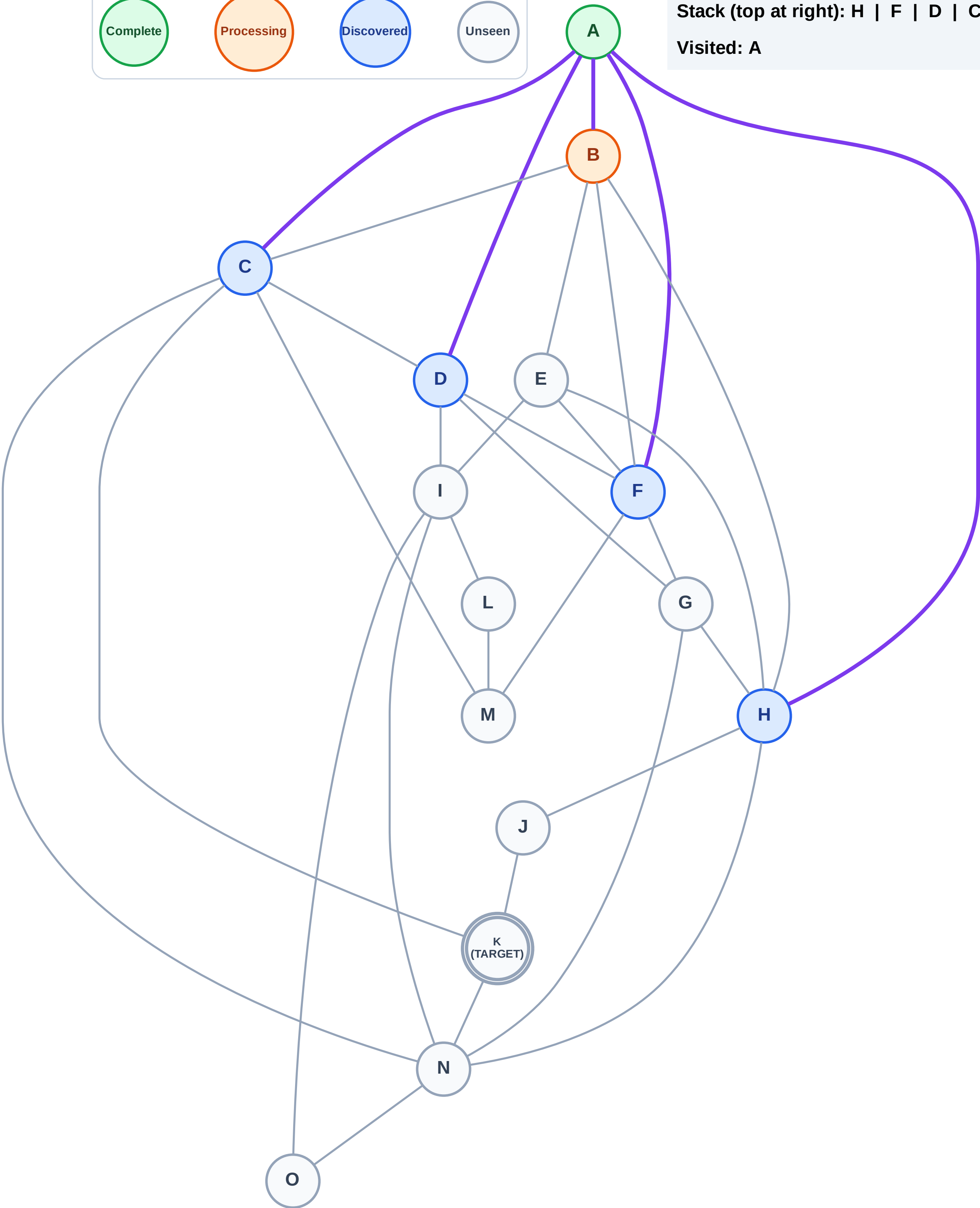
Processing

Discovered

Unseen

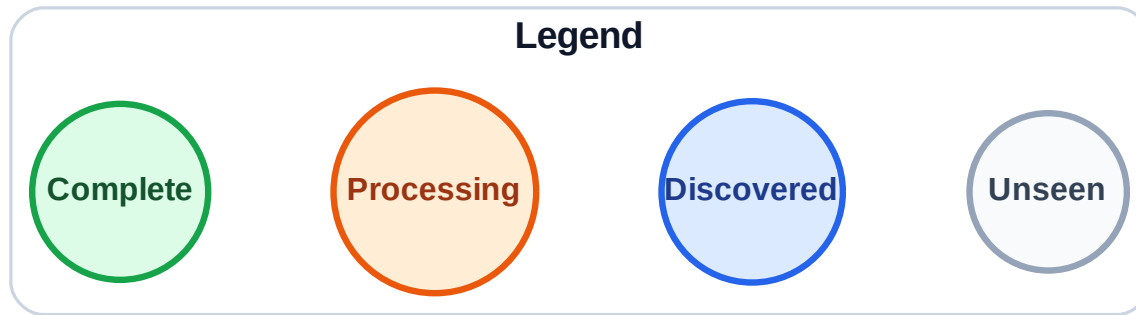
Stack (top at right): H | F | D | C

Visited: A

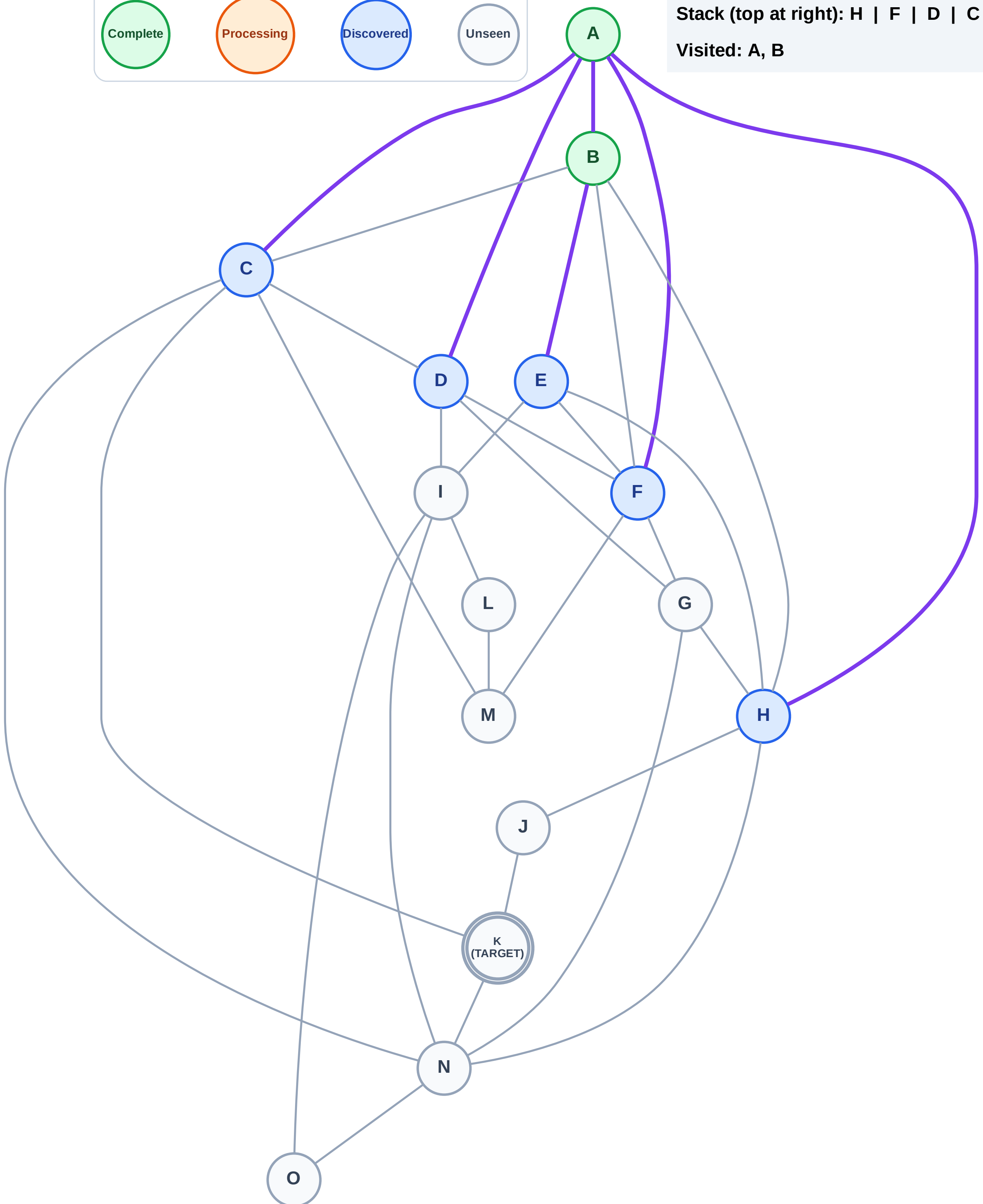


DFS Traversal

Step 5: Discover E from B



Stack (top at right): H | F | D | C | E
Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

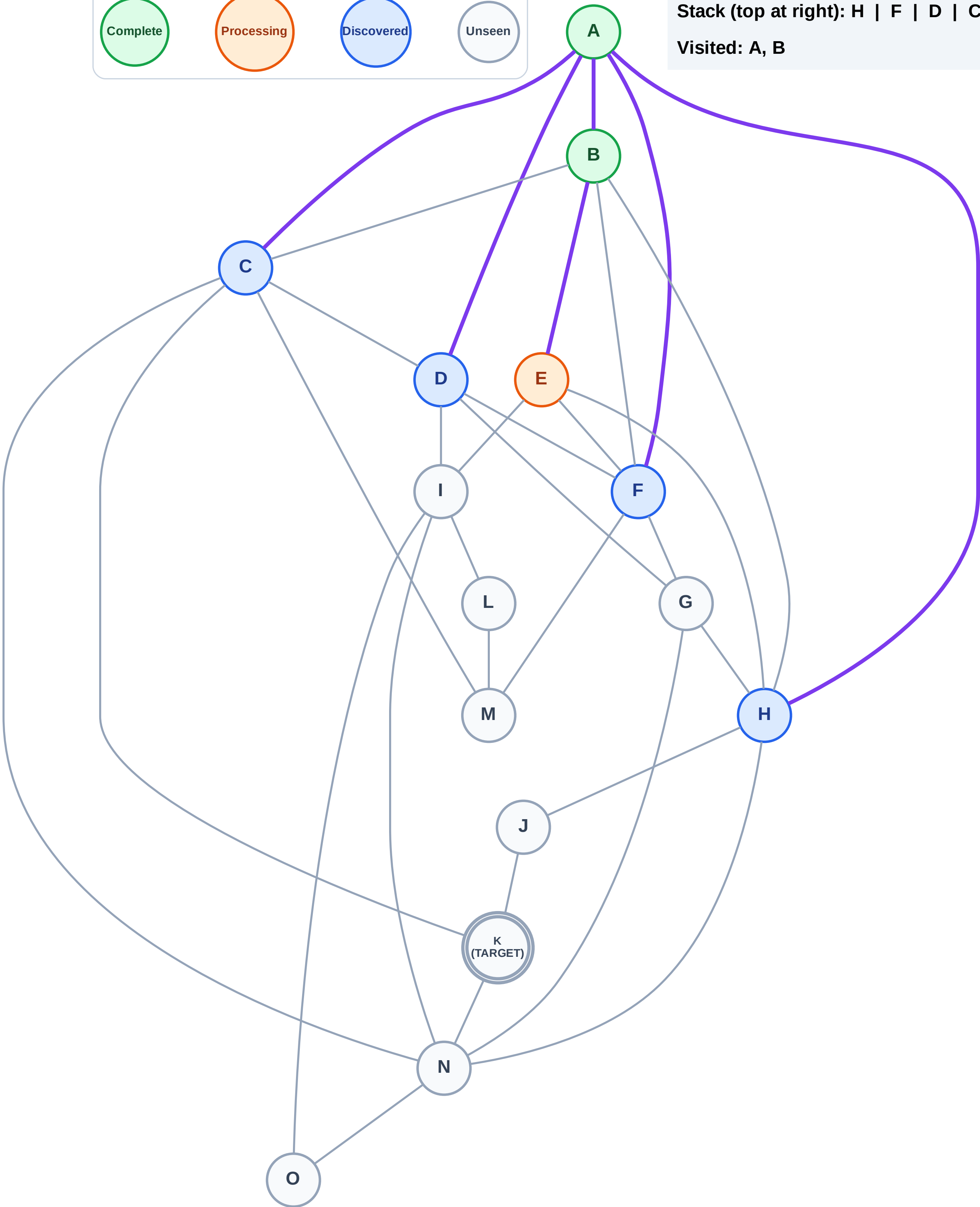
Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C
Visited: A, B



DFS Traversal

Step 7: Discover I from E

Legend

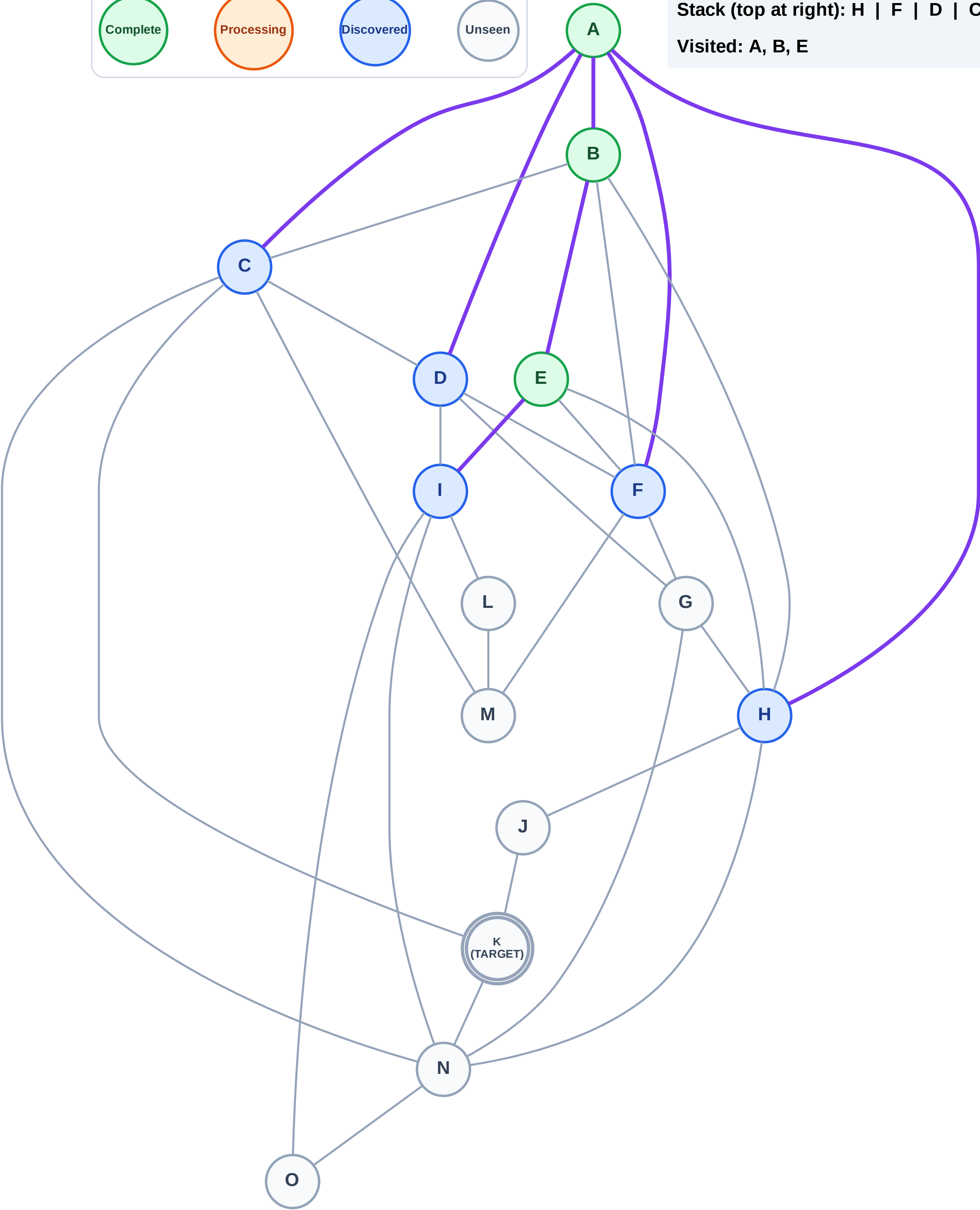
Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | I
Visited: A, B, E



DFS Traversal

Step 8: Process node I

Legend

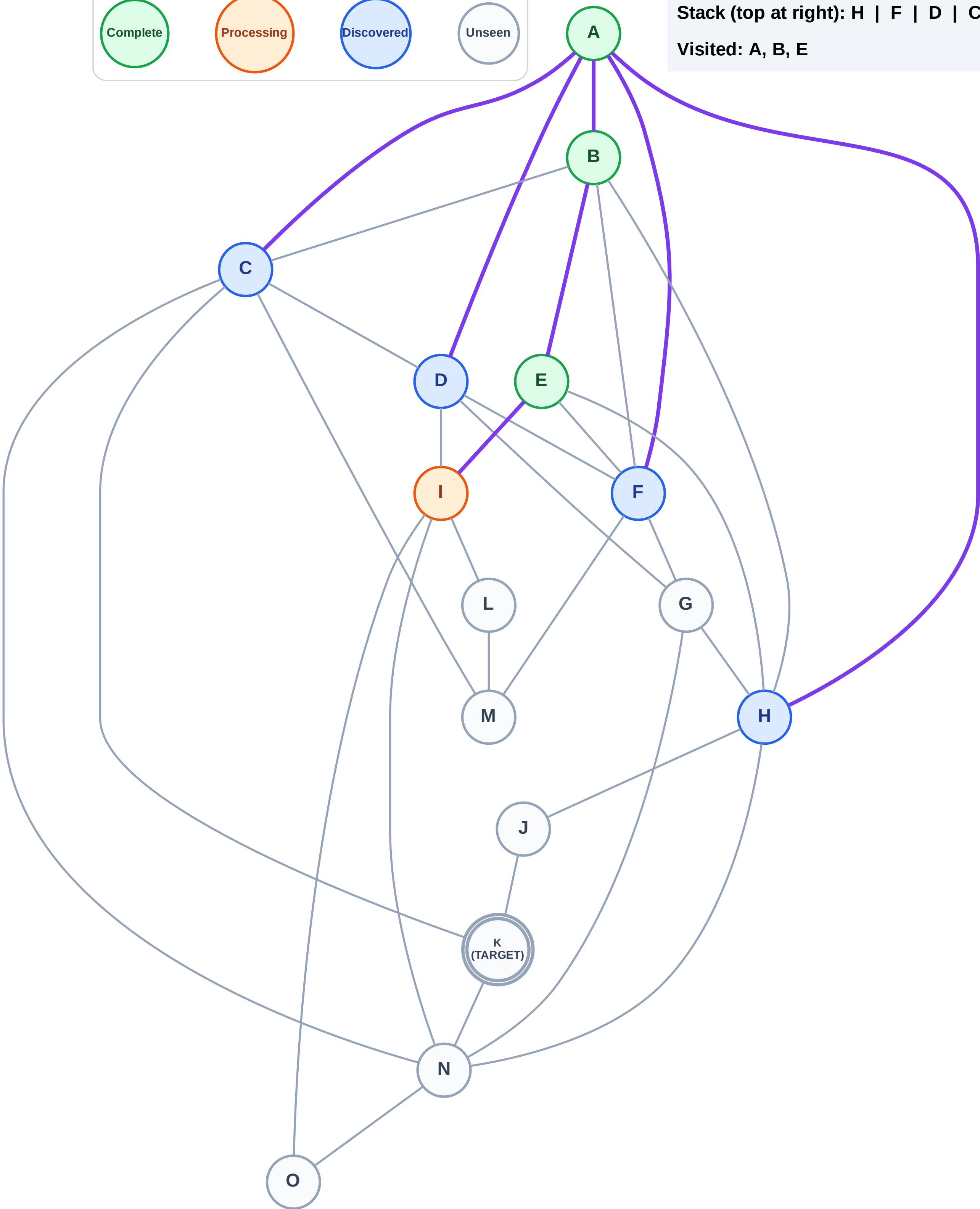
Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C
Visited: A, B, E



DFS Traversal

Step 9: Discover O, N, L from I

Legend

Complete

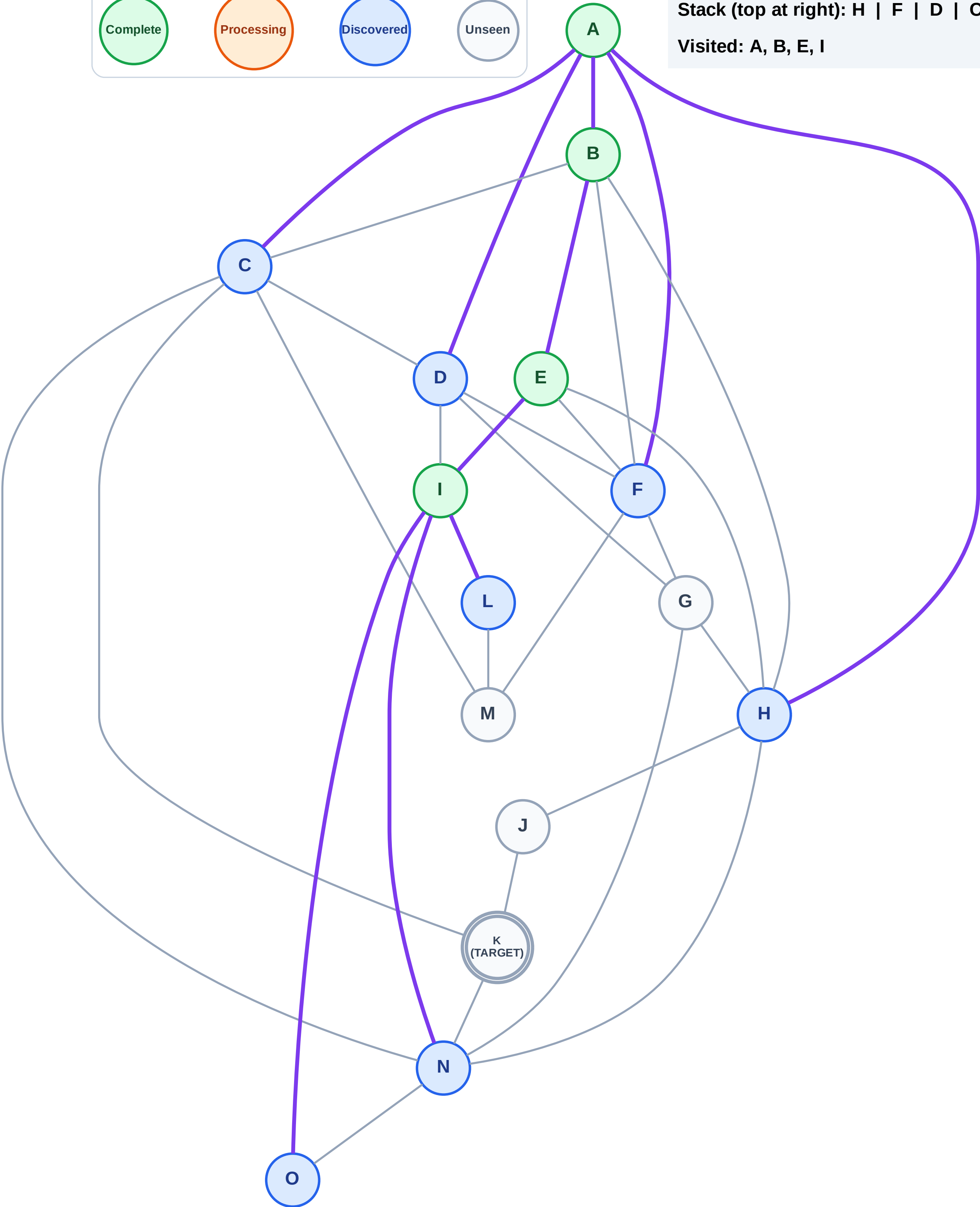
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N | L

Visited: A, B, E, I



DFS Traversal

Step 10: Process node L

Legend

Complete

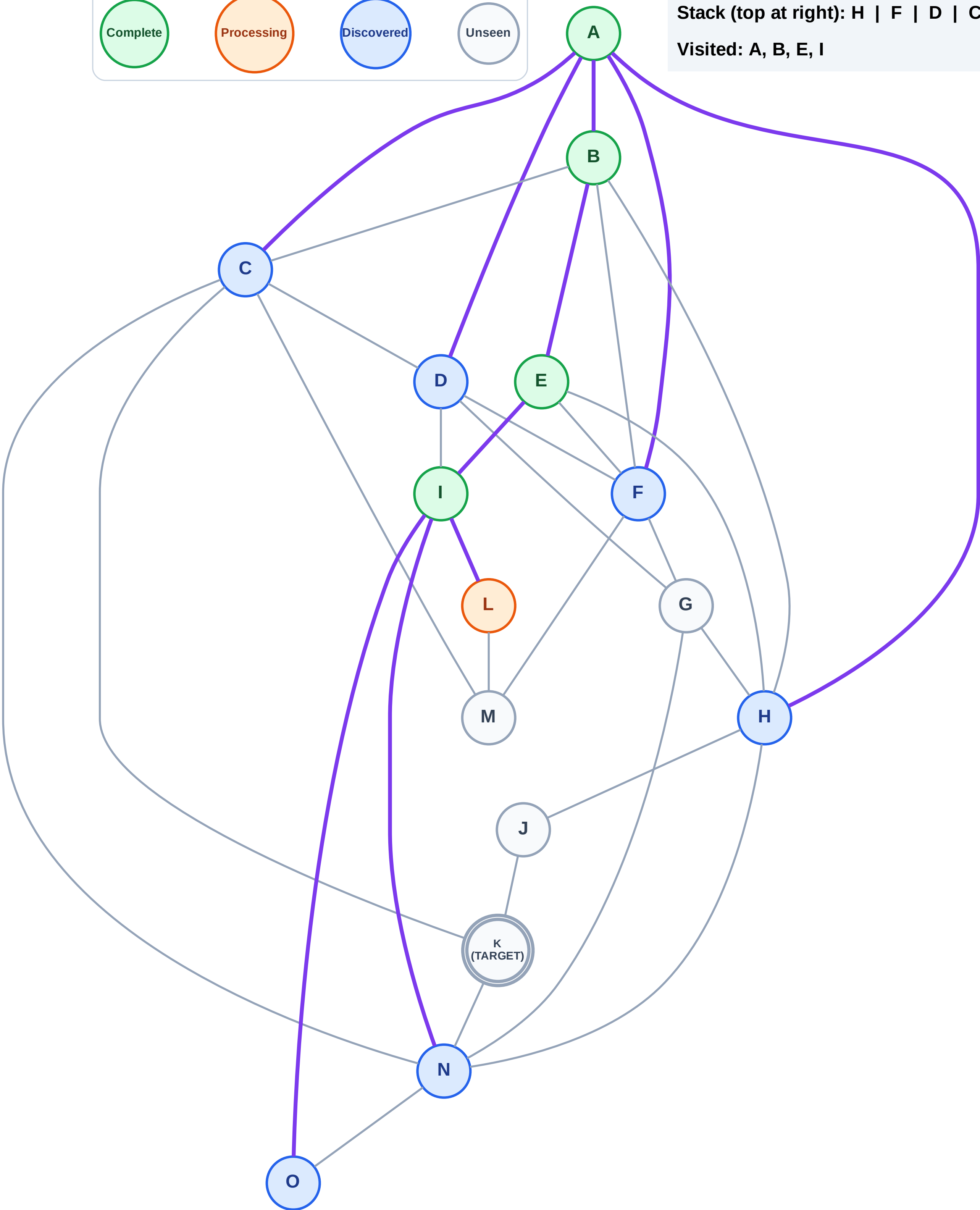
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N

Visited: A, B, E, I



DFS Traversal

Step 11: Discover M from L

Legend

Complete

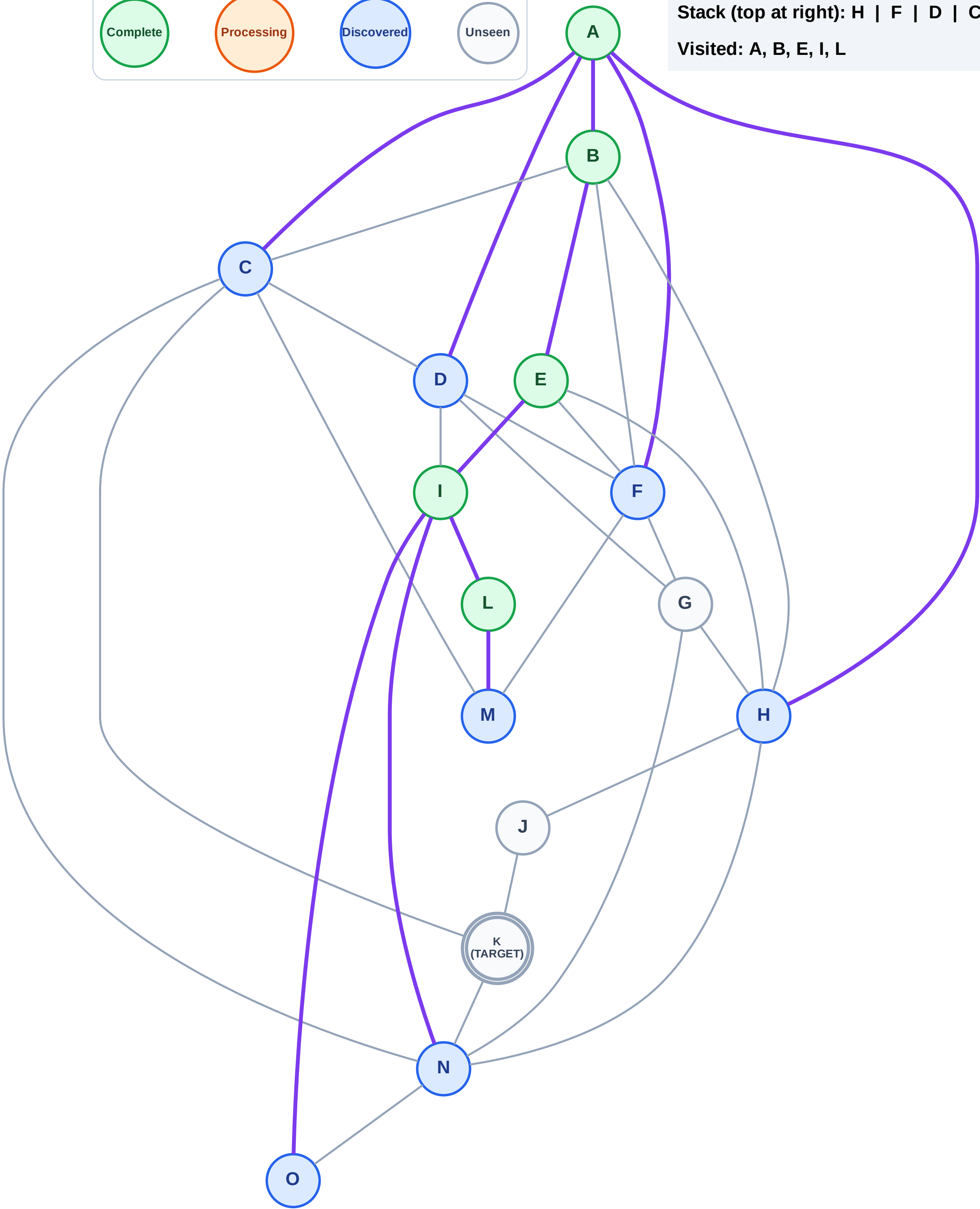
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N | M

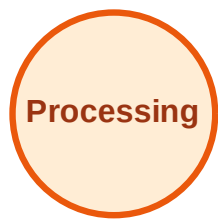
Visited: A, B, E, I, L



DFS Traversal

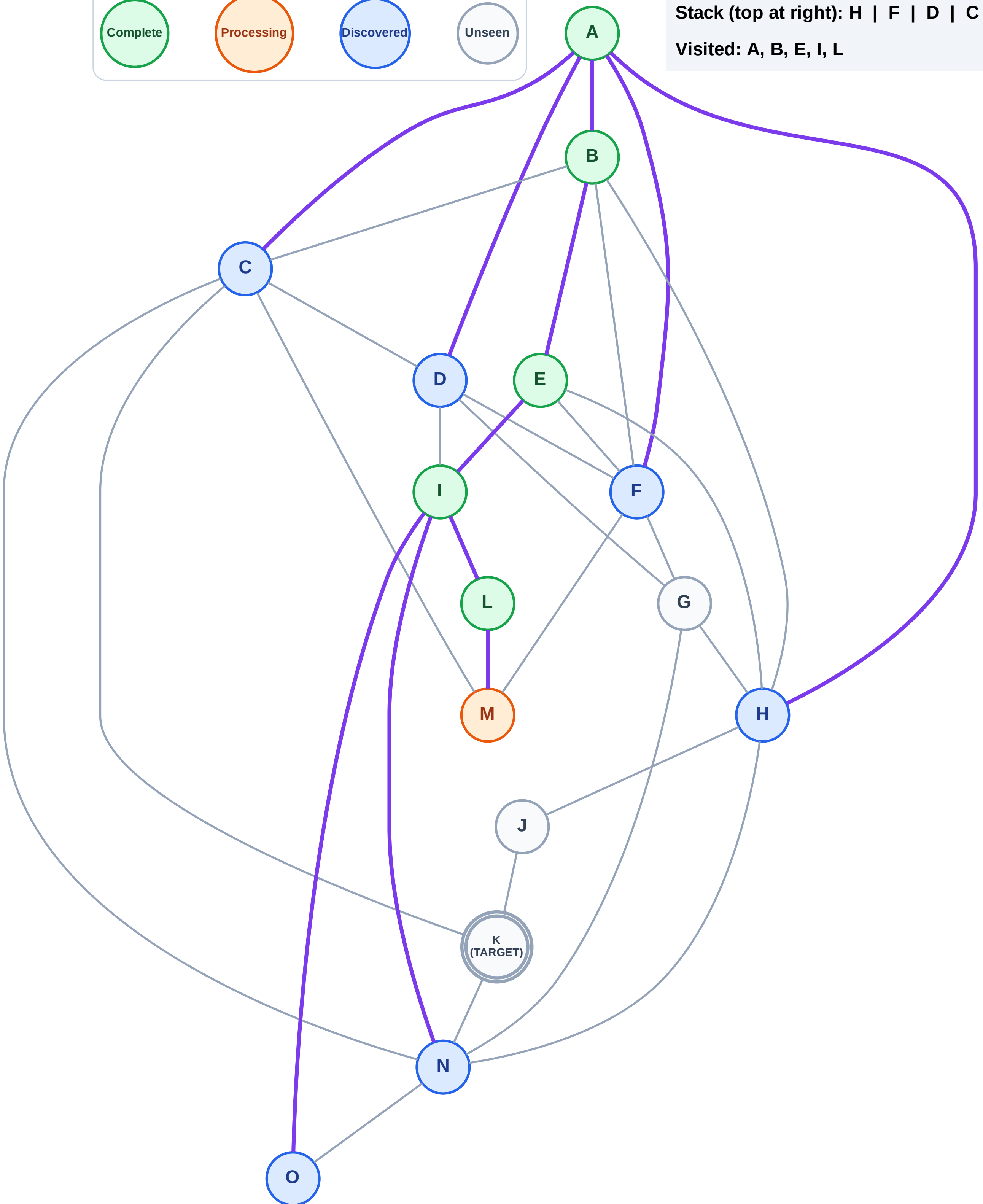
Step 12: Process node M

Legend



Stack (top at right): H | F | D | C | O | N

Visited: A, B, E, I, L



DFS Traversal

Step 13: No new neighbors from M

Legend

Complete

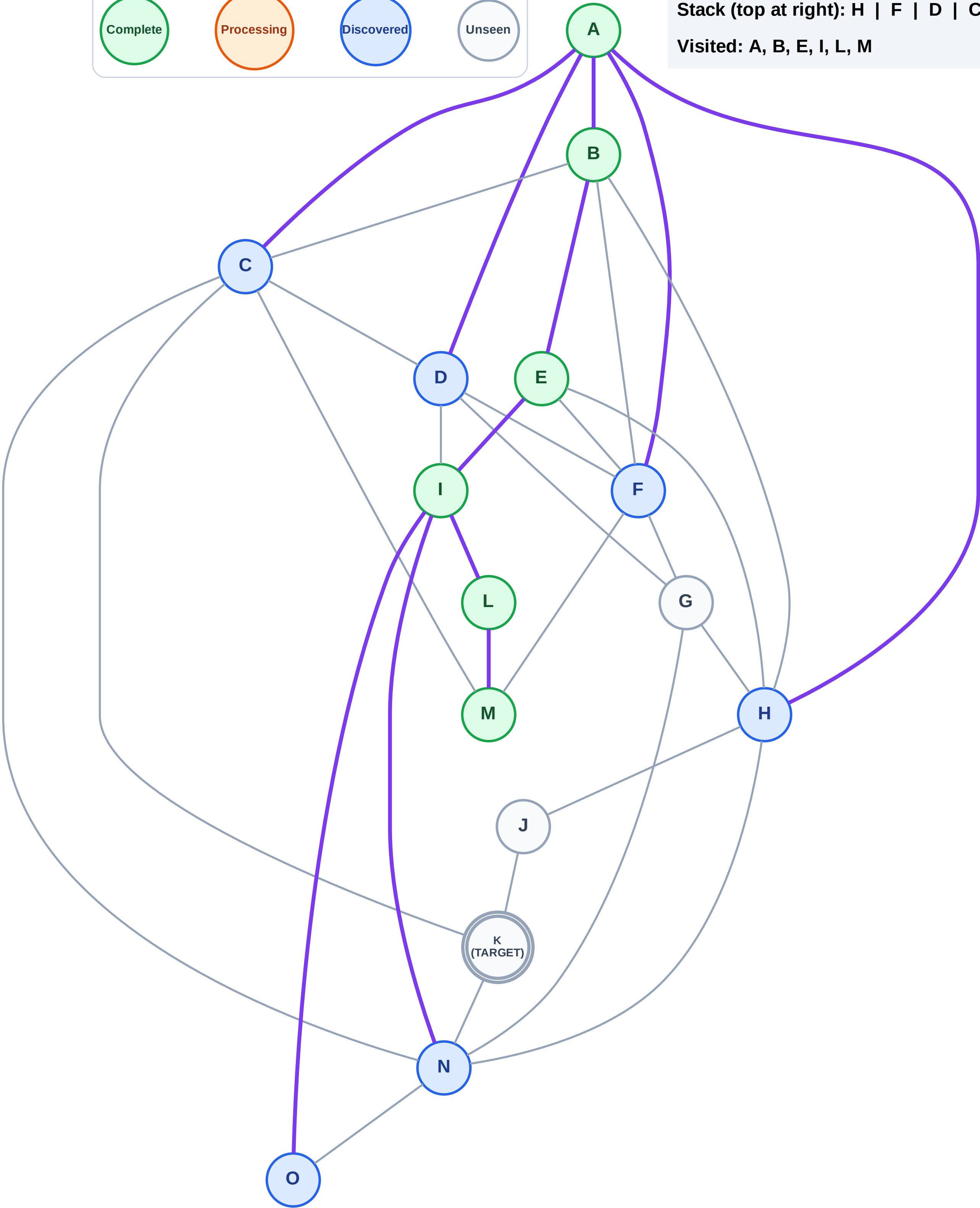
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O | N

Visited: A, B, E, I, L, M



DFS Traversal

Step 14: Process node N

Legend

Complete

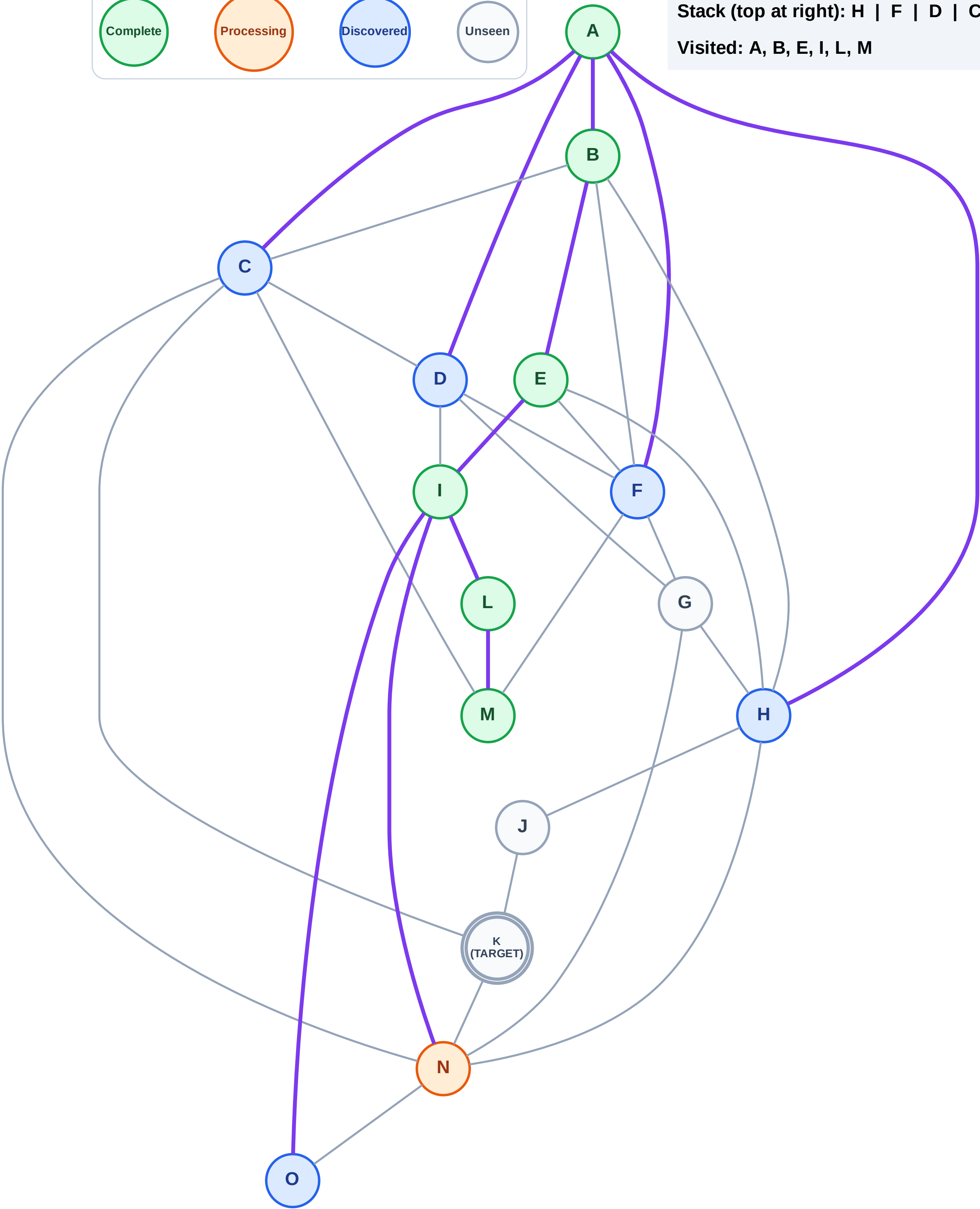
Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O

Visited: A, B, E, I, L, M



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → E → I → N → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): H | F | D | C | O
Visited: A, B, E, I, K, L, M, N

